Statistics 5810/6810: Data frames, Lists, Matrices

AND the Apply Family

Data Frame

- *Ordered* container of vectors
- Vectors must all be the *same length*
- Vectors can be *different types*

Reading data into R

- Many data sets are stored in text files.
- The easiest way to read these into R is using either the **read.table** or **read.csv** function, both of which return a data frame.
- There are quite a few options that can be changed. Some of the important ones are
  - file - name or URL
  - header - are column names at the top of the file?
  - sep - what divides elements of the table
  - na.strings - symbol for missing values, like 9999
  - Skip - number of lines at the top of the file to ignore

Wireless Data

- There are 5 wireless access points in a building
- A laptop emits a signal and the strength of the signal is recorded as each access point.
- Also recorded is the location of the laptop in the building.
- Measurements are taken at 254 locations

```
w =
read.table("http://www.math.usu.edu/~symanzik/teaching/
2012_stat5810/LectureNotes/wireless.txt", header=TRUE)
```

Helpful Functions for finding information out about the data frame

class(w)

[1] "data.frame"

names(w)

[1] "x" "y" "S1" "S2" "S3" "S4" "S5"

dim(w)

[1] 259 7

head(w)

```
    x   y  S1  S2  S3  S4  S5
1 225.0 144 -92 -78 -49 -92 -92
2   0.0 144 -75 -92 -87 -47 -92
3 111.0 132 -92 -92 -80 -70 -65
4 110.7 132 -87 -92 -67 -67 -62
5 105.0 132 -92 -92 -79 -66 -61
6  99.0 132 -92 -92 -79 -70 -61
```

summary(w$S1)

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
-92.00  -90.00  -80.00  -77.41  -71.00  -33.00
```

w[w$x > 200, "S2"]

```
 [1] -78 -74 -35 -43 -46 -43 -41 -48 -68 -71 -67 -70 -58 -61
[15] -64 -73 -71 -67 -68 -61 -60 -57 -58 -56 -48 -51 -68 -54
[29] -51 -48 -46 -35 -67
```

We subset rows and columns of data frames

We subset by **position**, **exclusion, logical, name**, and **all**

Lists

- Data frames are actually a special kind of *list*.
- Unlike a data frame each element in a list can have a different length.
- Actually, each element can be either a list, data frame, vector, matrix …
- Note that the elements are not associated with one another by position, as they were in a given row of a data frame.

Rainfall

- Daily rainfall collected for 5 weather stations in Colorado
- **rain** is a list of length 5
  - One element for each station
  - Each element is a numeric vector of rain measurements
  - Stations not in operation for the same length of time
- **day** has the same structure as rain

```
load(url("http://www.math.usu.edu/~symanzik/teaching/2012_stat5810/LectureNotes/rainfallCO.rda"))
class(rain)
[1] "list"
length(rain)
[1] 5
names(rain)
[1] "st050183" "st050263" "st050712" "st050843" "st050945"
```

Indexing lists

- Lists can be indexed by name, using $.

```
class(rain$st050183)
[1] "numeric"
length(rain$st050183)
[1] 9878
head(rain$st050183)
[1]  0 10 11  1  0  0
```

Indexing lists

- Lists can also be indexed like vectors, using []. The result will be another list.

```
class(rain["st050183"])
```
[1] "list"
```
length(rain["st050183"])
```
[1] 1

Indexing lists

- To extract individual elements of a list, enclose the index in [[]]. The result will an object of the same type as the element of the list.

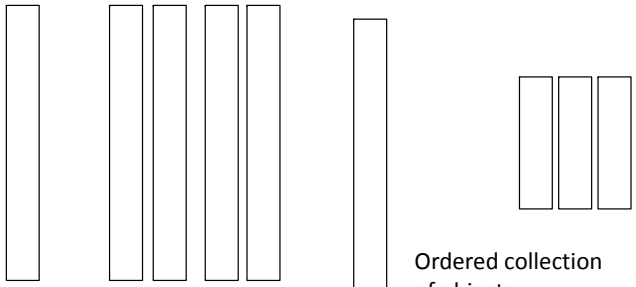
```
class(rain[[1]])
```
[1] "numeric"
```
head(rain[[1]])
```
[1] 0 10 11 1 0 0

Reminder: Summary of Data Structures

Types of structures

- To summarize, the data structures we have encountered so far are:
  - vector
  - data frame
  - list
  - matrix
- Matrices and arrays are actually just stored as vectors with shape information, so our discussions of "vectorized" calculations hold for matrices as well.
- This is NOT true for lists and data frames.

Vector – Data frame - List



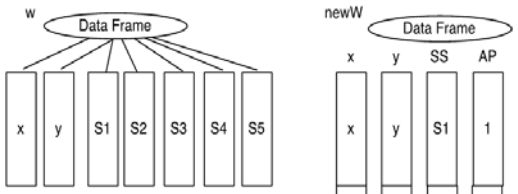
Ordered collection of vectors all same length

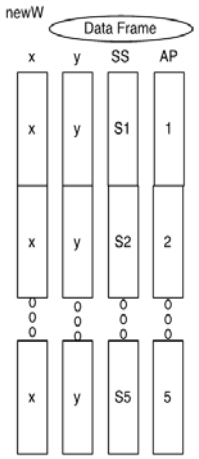
Ordered collection of objects

Return to the Wifi problem

We will reformat the data frame into another data frame

Revise the Structure






```
X = rep(w$x, 5)
Y = rep(w$y, 5)
AP = rep(1:5, each = nrow(w))
SS = c(w$S1, w$S2, w$S3, w$S4, w$S5)

newW = data.frame(x = X, y = Y, AP, SS)
head(newW)
   x y AP SS
1 225.0 144 1 -92
2 0.0 144 1 -75
3 111.0 132 1 -92
4 110.7 132 1 -87
5 105.0 132 1 -92
6 99.0 132 1 -92
```

Apply Functions

- Sometimes we want an operation to be applied to each element of a list, to each vector in a data frame, or to individual dimensions of a matrix
- R provides the *apply* mechanism to do this.
- There are several apply functions:
  - sapply() and lapply() for lists and data frames
  - apply() for matrices
  - tapply() for "tables", i.e. ragged arrays as vectors

- With these functions we can avoid looping, and instead we write code that is meaningful in a statistical setting.
- For example with our list of rainfall data, each element represents the measurements taken at a particular weather station and when we think about studying the average rainfall at each station - we don't think in terms of loops.

lapply() and sapply()

- The **lapply** and **sapply** both apply a specified function to each element of a list.
- The former returns a list object and the latter a vector when possible.

Mean rainfall at each station

lapply(rain, mean)

```
$st050183
[1] 6.631707

$st050263
[1] 3.798993

$st050712
[1] 5.102299

$st050843
[1] 6.084607

$st050945
[1] 4.549296
```

sapply(rain, mean)

```
st050183 st050263 st050712
6.631707 3.798993 5.102299

st050843 st050945
6.084607 4.549296
```

Additional arguments

lapply(rain, mean, na.rm = TRUE, trim = 0.1)

```
$st050183
[1] 2.393978

$st050263
[1] 0.9875949

$st050712
[1] 0.7895235

$st050843
[1] 1.238481

$st050945
[1] 0.7366283
```

args(lapply)

function (X, FUN, ...)

X takes the list object

FUN is the function to apply to each element in X

… allows any number of arguments to be passed to FUN

tapply()

- This function is useful to apply a function to subsets of a vector.

x = 1:10

[1] 1 2 3 4 5 6 7 8 9 10

v = c(1, 1, 1, 0, 0, 0, 1, 1, 1, 0)

[1] 1 1 1 0 0 0 1 1 1 0

tapply(x, v, mean)

0 1

6.25 5.00

tapply(x, v, median)

0 1

5.5 5.0

compare with

mean(x[v == 0])

[1] 6.25

mean(x[v == 1])

[1] 5

Now it's your turn to try it out

Rainfall data

- Maximum rainfall at each station

sapply(rain, max)

- 99th percentile of rainfall at each station

sapply(rain, quantile, probs = 0.99)

Rainfall data

- Check that the number of recordings at each station matches the number of days recorded at the corresponding station

all(sapply(rain, length) == sapply(day, length))

Rainfall data

- Number of years each station is in operation

Year = lapply(day, floor)
Uyear = lapply(Year, unique)
OpYear = sapply(Uyear, length)
For any one station:
length(unique(floor(day[[1]])))
sapply(day, length(unique(floor(?))))

sapply(day, function(x) length(unique(floor(x))))

- Proportion of days it rained at each station

sapply(rain, function(x)
sum(x > 0)/length(x))

Web Caching

Web Caching

- When you use a search engine to look for a Web page, the search engine looks through its cache.
- The cache is created by regularly crawling the web and taking copies of every page that has changed since the last time it visited the site.
- It's time consuming and costly to crawl the web.
- How often do you need to crawl the web in order to keep its stock fresh for the search engine?

Web Caching

- Data: a collection of web sites were visited regularly over a period of time to see whether or not they had changed.
- Question: Given a new website, how often would you recommend as the length of time between visiting the site?

```
load(url("http://www.math.usu.edu/~symanzik/teaching/2012_stat5810/LectureNotes/refreshSite.rda"))
```

```
class(refreshSite2)
[1] "list"
length(refreshSite2)
[1] 10000

class(refreshSite2[[1]])
[1] "integer"
length(refreshSite2[[1]])
[1] 13
length(refreshSite2[[2]])
[1] 3
```

refreshSite2[1]

[[1]]

[1] 35 134 155 157 177 204 314 315 319 350 366 369 371

The first web site changed 13 times.

The first change was observed on the 35th visit, the second on the 134th visit, etc

refreshSite2[c(1, 2, 4, 6)]

```
[[1]]
 [1]  35 134 155 157 177 204 314 315 319 350 366 369 371

[[2]]
[1] 552 604 672

[[3]]
 [1]  30  36  45  65  88 154 157 166 169 197 199 204 220 223 225 235 250
[18] 251 253 311 319 321 341 346 365 367 369

[[4]]
 [1]   4   5  28  52  76 100 116 129 155 196 221 318 356 438 553 562 609
```

Extracting information

- We can think of these values as units of time, e.g. if the visits occurred daily then 5 is 5 days.
- Some characteristics of interest to us are:
  - The number of changes over the observed time period
  - The length of "time" between changes
  - The times at which a change was observed

Example: One Web site

[1] 35 134 155 157 177 204 314 315 319 350 366 369 371

- The number of changes over the observed time period: 13
- The length of "time" between changes:

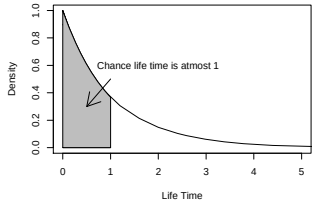
35, 99, 21, 2, 20, 27, 110, 1, 4, 31, 16, 3, 2

- The times at which a change was observed

These are the values given to us: 35, 134, …

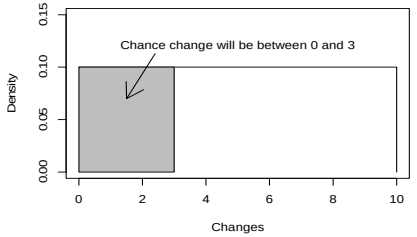
Properties of the time between

Might behave like a waiting time - the density curve for the time between changes looks like the exponential



Properties of the times

- Might behave like a uniform – the density curve for the locations might be uniform distribution (flat)



Cautions

- If the site changed more than once between visits, we wouldn't be able to tell
- We can't tell when the site changed, only that it changed at some point in the interval between visits

Matrices and Arrays

Matrices and Arrays

- Rectangular collection of elements
- Dimensions are two, three, or more
- Homogeneous primitive elements (e.g. all numeric or all character)

- You can create a matrix in R using the **matrix** function.
- By default, matrices in R are assigned by *column-major* order.
- You can assign them by *row-major* order by setting the byrow argument to TRUE. Note that the first argument to matrix is a vector, so all elements must be of the same type (numeric, character, or logical).

```
m = matrix(1:6, nrow = 2)
     [,1] [,2] [,3]
[1,]    1    3    5
[2,]    2    4    6
```

Arrays – matrices in higher dimensions

```
x = array(1:30, c(4, 3, 2))
x
, , 1
     [,1] [,2] [,3]
[1,]    1    5    9
[2,]    2    6   10
[3,]    3    7   11
[4,]    4    8   12

, , 2
     [,1] [,2] [,3]
[1,]   13   17   21
[2,]   14   18   22
[3,]   15   19   23
[4,]   16   20   24
```

- The integers 1, 2, … , 30 are arranged in a 3-dimensional array
- The array has:
  - 4 rows
  - 3 columns
  - 2 panels

```
x[1:2, 3, 2]
[1] 21 22

x[ , 2, 1]
[1] 5 6 7 8

x[3:4, c(3, 1), 1]
     [,1] [,2]
[1,]   11    3
[2,]   12    4
```

apply()

- **apply(m, margin, sum)** for the matrix m, a function (here the sum) is applied. A margin of 1 applies the function to the rows, and a margin of 2 applies it to the columns.

```
m = matrix(1:6, nrow = 2)
     [,1] [,2] [,3]
[1,] 1    3    5
[2,] 2    4    6
apply(m, 1, sum)
[1] 9 12
apply(m, 2, sum)
[1] 3 7 11
```